2) Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df = pd.DataFrame({
    'Date': ['2025-02-01', '2025-02-02', '2025-02-03', '2025-02-04', '2025-02-05',
             '2025-02-06', '2025-02-07', '2025-02-08'],
    'Product': ['Monitor', 'Keyboard', 'Mouse', 'Monitor', 'Keyboard',
                'Mouse', 'Monitor', 'Keyboard'],
    'Sales': [28000, 12000, 8000, 35000, 15000, 9500, 42000, 18000],
    'Quantity': [4, 10, 12, 5, 12, 15, 6, 14],
    'Region': ['East', 'West', 'North', 'South', 'East',
               'North', 'West', 'South']
})
print(df.head())
print(df.isnull().sum())
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
print(df.describe())
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print(product_summary)
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
```

```
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date')['Sales'].sum().reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'], marker='o')
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.show()
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)
print(pivot_table)
correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)
plt.figure(figsize=(8, 6))
plt.imshow(correlation_matrix, cmap='coolwarm')
plt.colorbar()
plt.xticks(range(len(correlation_matrix.columns)), correlation_matrix.columns)
plt.yticks(range(len(correlation_matrix.columns)), correlation_matrix.columns)
plt.title('Correlation Matrix')
```

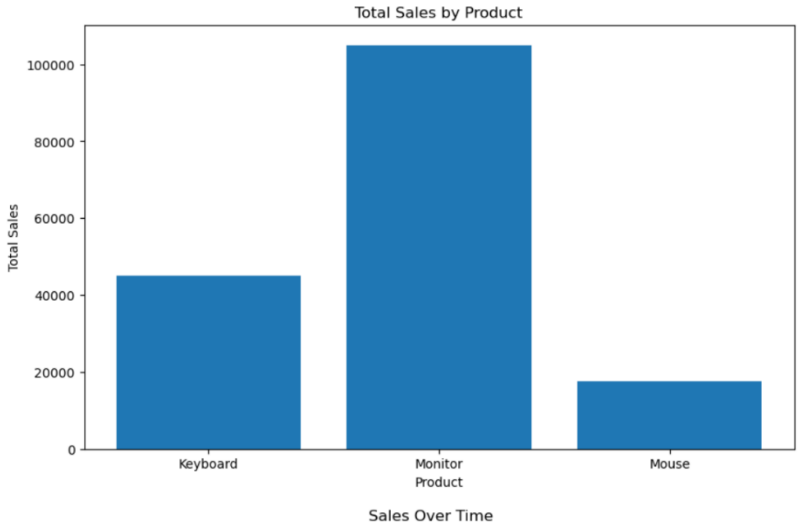
plt.show()

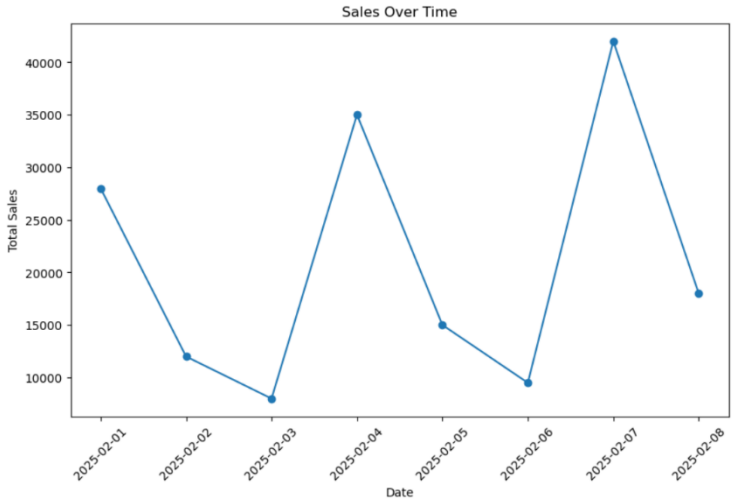
Output:

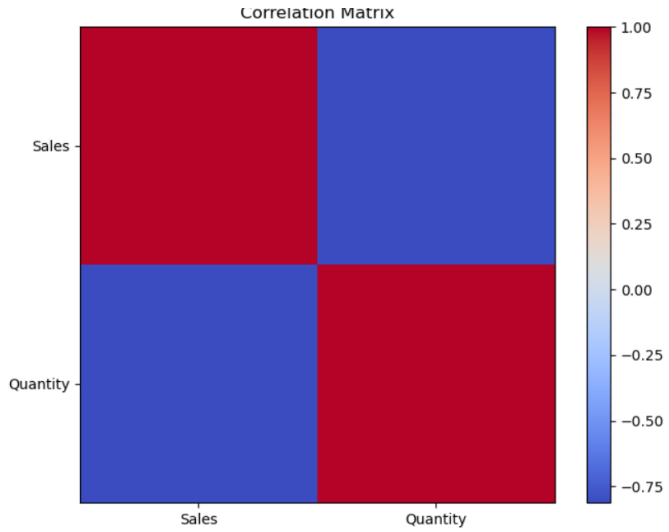
```
         Date   Product  Sales  Quantity Region
0  2025-02-01   Monitor  28000         4   East
1  2025-02-02  Keyboard  12000        10   West
2  2025-02-03     Mouse   8000        12  North
3  2025-02-04   Monitor  35000         5  South
4  2025-02-05  Keyboard  15000        12   East
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
              Sales   Quantity
count      8.000000   8.000000
mean   20937.500000   9.750000
std    12610.759295   4.234214
min     8000.000000   4.000000
25%    11375.000000   5.750000
50%    16500.000000  11.000000
75%    29750.000000  12.500000
max    42000.000000  15.000000
    Product   Sales  Quantity
0  Keyboard   45000        36
1   Monitor  105000        15
2     Mouse   17500        27
```

Product	Keyboard	Monitor	Mouse
Region			
East	15000	28000	0
North	0	0	17500
South	18000	35000	0
West	12000	42000	0

	Sales	Quantity
Sales	1.000000	-0.812317
Quantity	-0.812317	1.000000







Result:

The dataset in CSV format and performed data processing and visualised was analysed.